Connection: $T_uP = V_u \oplus H_u$ $\boxed{V_u = \ker(d\pi_u)}$ $\pi : P \to M$

$dR_g|_u (H_u) \subseteq H_u$ $R_g : u \in P \mapsto u \triangleleft g \in P$ define $\omega$

let $X \in T_uP$ $\Rightarrow$ then is unique $A \in \mathfrak{g}$ s.t. $\mathrm{proj}_{V_u}(X) = A^*_u$

$\omega_u(X) = A$

equivalent approach: connection 1-form ($\mathfrak{g}$-valued) $\omega$

$$\begin{cases} \omega_u : T_uP \to \mathfrak{g} \\ \omega_u(A^*_u) = A \quad \text{for } A \in \mathfrak{g} \\ R_g^* \omega = g^{-1} \omega g \end{cases}$$

$H_u = \ker(\omega_u)$

$A^*_u$ → fundamental vector field

$A^*_u f = \frac{d}{dt}\Big|_{t=0} f(u \triangleleft e^{tA})$

$A^*$ is vertical

$d\pi_u(A^*_u) = 0$

$\hookrightarrow d\pi_u(A^*_u) f = A^*_u (f \circ \pi)$

$= \frac{d}{dt}\Big|_{t=0} f(\pi(u \triangleleft e^{tA}))$

$= \frac{d}{dt}\Big|_{t=0} f(\pi(u)) = 0$

$\boxed{\text{local description of } \omega}$

local sections of $P$

assume we have local trivialisation $\{(U_\alpha, \psi_\alpha)\}$

$\psi_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times G$ $(\mathrm{proj}_1 \circ \psi_\alpha = \pi$

transition functions $\psi_{\alpha\beta} = \psi_\alpha \circ \psi_\beta^{-1} : U_\alpha \cap U_\beta \times G \to U_\alpha \cap U_\beta \times G$

$(u, g) \mapsto (u, t_{\alpha\beta}(u) \triangleleft g)$ clutching function

$\in G$

$t_{\alpha\beta} : U_\alpha \cap U_\beta \to G$

require that $\mathrm{proj}_2\, \psi_{\alpha\beta}(u, g) = g \triangleleft t_{\alpha\beta}(u)$

fundamental sections of $P$ (local)

$\pi(\sigma_\alpha(x)) = \pi \circ \psi_\alpha^{-1}(x, e) = x$

$\sigma_\alpha : U_\alpha \subseteq M \to P$ s.t. $\pi \circ \sigma_\alpha = id$

$\sigma_\alpha(x) := \psi_\alpha^{-1}(x, e)$ $x \in U_\alpha$ $(x, e) \in U_\alpha \times G \xmapsto{\psi_\alpha^{-1}} \pi^{-1}(U_\alpha) \subseteq P$

if $U_\alpha \cap U_\beta \neq \emptyset$

$$\sigma_\beta(x) = \psi_\beta^{-1}(x, e) = \psi_\alpha^{-1} \circ \psi_{\alpha\beta}(x, e) = \psi_\alpha^{-1}(x, t_{\alpha\beta}(x) \triangleright e)$$
$$= \psi_\alpha^{-1}(x, t_{\alpha\beta}(x))$$

$u \in \pi^{-1}(U_\alpha)$ $\qquad u \triangleleft g = \psi_\alpha^{-1}(\psi_\alpha(u) \triangleleft g)$

$\sigma_\alpha(x) = \psi_\alpha^{-1}(x, e)$

$\Rightarrow (x, e) = \psi_\alpha(\sigma_\alpha(x))$

$$\psi_\alpha^{-1}(x, e \triangleleft t_{\alpha\beta}(x)) = \psi_\alpha^{-1}((x, e) \triangleleft t_{\alpha\beta}(x))$$
$$= \psi_\alpha^{-1}(\psi_\alpha(\sigma_\alpha(x)) \triangleleft t_{\alpha\beta}(x))$$
$$= \sigma_\alpha(x) \triangleleft t_{\alpha\beta}(x)$$

$$\boxed{\sigma_\beta(x) = \sigma_\alpha(x) \triangleleft t_{\alpha\beta}(x)}$$ $\rightarrow$ gauge transformation

---

can use the $\{\sigma_\alpha\}$ sections to replace $\omega$ (1-form on $P$) with a collection of local $\mathfrak{g}$-valued 1-forms on $M$

section 1-form

$\omega_\alpha := \sigma_\alpha^* \omega \qquad \sigma_\alpha : U_\alpha \to P$

$\omega_\alpha|_x : T_x U_\alpha = T_x M \to \mathfrak{g}$ $\qquad$ 1-form on $U_\alpha \subseteq M$

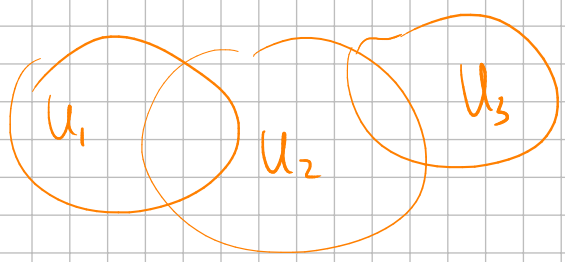
$$\omega_\alpha|_x(v) = (\sigma_\alpha^* \omega)_x(v) = \omega_{\sigma_\alpha(x)}(\underbrace{d\sigma_\alpha|_x(v)}_{\in T_{\sigma(x)}P})$$

---

transformation law

gauge transformation

$$\boxed{\omega_\beta = t_{\alpha\beta}^{-1} \omega_\alpha t_{\alpha\beta} + t_{\alpha\beta}^{-1} d t_{\alpha\beta}}$$




---

$f : M \to N$ smooth

differential of $f$ at $a \in M$.
(pushforward) $\Rightarrow df_a : T_a M \to T_{f(a)} N$

if $X \in T_a M$ and $F \in \mathcal{C}^\infty(N, \mathbb{R})$

define $df_a(X) F = X(\underbrace{F \circ f}_{M \to \mathbb{R}})$

---

pullback: map covariant tensor fields on $N$ to covariant tensor fields on $M$

$f^* : \mathcal{C}^\infty(N, \mathbb{R}) \to \mathcal{C}^\infty(M, \mathbb{R})$
$F \mapsto F \circ f$

for 1-forms

$f^* : \Lambda^1(N) \to \Lambda^1(M)$

$\omega \in \Lambda^1(N) \qquad a \in M, \ v \in T_a M$

$(f^* \omega)_a : T_a M \to \mathbb{R}$
$v \mapsto \omega_{f(a)}(df_a(v))$

$\Downarrow$

right action of $G$ on $\mathfrak{g}$:

$X \in \mathfrak{g} \quad g \in G \qquad X \triangleleft g = g^{-1} X g \in \mathfrak{g}$

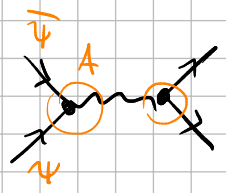
$(X \triangleleft g) \triangleleft h = (g^{-1} X g) \triangleleft h = h^{-1} g^{-1} X g h = X \triangleleft gh$

extend by using

$f^*(\alpha \otimes \beta) = f^*\alpha \otimes f^*\beta$

$f^*(d\alpha) = d f^*\alpha$

$f^*(\alpha \wedge \beta) = f^*\alpha \wedge f^*\beta$




ex 1 trivial $\quad M = \mathbb{R}^4$ (Minkowski) $\qquad P = M \times U(1)$

global trivialization $\quad \sigma : P \to P$ (identity) $\qquad \underline{u}(1) = i\mathbb{R}$

1 global-bal field : $\mathcal{A}$ is a $i\mathbb{R}$-valued 1-form

in coordinates: $\quad \mathcal{A} = \mathcal{A}_\mu dx^\mu \qquad \mathcal{A}_\mu : M \to i\mathbb{R}$

interpret $\quad A_\mu = i\mathcal{A}_\mu \quad$ as 4-potential $\qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ Faraday tensor

gauge transformation: $\quad g : M \to U(1)$

gauge transformation of $\mathcal{A}$: $\qquad \tilde{\mathcal{A}}_\mu(x) = g^{-1}(x) \mathcal{A}_\mu g(x) + g^{-1} \frac{\partial g}{\partial x^\mu}(x)$

cancel out

$= \mathcal{A}_\mu(x) + g^{-1} \frac{\partial g}{\partial x^\mu}(x)$

$g(x) = e^{i\theta(x)}$

$g^{-1} \frac{\partial g}{\partial x^\mu} = i e^{-i\theta} e^{i\theta} \frac{\partial \theta}{\partial x^\mu} = i \frac{\partial \theta}{\partial x^\mu}$

$\tilde{\mathcal{A}}_\mu = \mathcal{A}_\mu + i \frac{\partial \theta}{\partial x^\mu}$

$\tilde{A}_\mu = i\tilde{\mathcal{A}}_\mu = i\left(\mathcal{A}_\mu + i \frac{\partial \theta}{\partial x^\mu}\right) = \underbrace{A_\mu - \frac{\partial \theta}{\partial x^\mu}}$

$\to$ EM gauge transformation

$\tilde{F}_{\mu\nu} = F_{\mu\nu}$

② <u>Magnetic monopole</u> no electric field, static magnetic field

$\nabla \cdot B = q\rho$ ← monopole $\qquad \rho = \delta^3(x)$

$M = \mathbb{R}^3 \setminus \{0\}$ $\qquad B$ 2-form on $\mathbb{R}^3$

$$B = B_x\, dy\wedge dz + B_y\, dz\wedge dx + B_z\, dx\wedge dy$$

$$dB = \partial_x B_x\, dx\wedge dy\wedge dz + \partial_y B_y\, dy\wedge dz\wedge dx + \partial_z B_z\, dz\wedge dx\wedge dy$$

$$= (\partial_x B_x + \partial_y B_y + \partial_z B_z)\, dx\wedge dy\wedge dz$$

$$\rightarrow \quad dB = \rho\, dx\wedge dy\wedge dz$$

to make sense of this we integrate

$$\int_{\mathbb{R}^3\setminus\{0\}} \nabla\cdot B\, f\, d^3x = q f(0) \qquad \rightarrow \text{find } B \text{ that works}$$

$$B = \frac{q}{4\pi}\frac{\vec{r}}{\|r\|^3} \qquad \nabla\cdot B = \frac{q}{4\pi}\left(\frac{3}{\|r\|^3} + \vec{r}\cdot\nabla\frac{1}{\|r\|^3}\right)$$

$$\nabla\cdot(\|r\|^2)^{-3/2} = -\frac{3}{2}\|r\|^{-2\cdot\frac{1}{2}}\,\nabla\cdot\|r\|^2$$

$$= \frac{3}{2}\|r\|\left(\partial_x(x^2+y^2+z^2),\ \partial_y(\text{—}),\ \partial_z(\text{—})\right)$$

$$= 3\frac{\vec{r}}{\|r\|}$$

<u>goal:</u> write $B = dA$ (find vector potential)

<u>as 2-form:</u> $$B = \frac{q}{4\pi\|\vec{x}\|^3}\left(x\, dy\wedge dz + y\, dz\wedge dx + z\, dx\wedge dy\right)$$

B <u>not exact</u>

locally we can have 2 patches that work

$U_+ = \mathbb{R}^3 \setminus \{(0,y) \mid y > 0\}$ $\qquad U_+ \cup U_- = \mathbb{R}^3 \setminus \{0\}$

$U_- = \mathbb{R}^3 \setminus \{(0,y) \mid y < 0\}$ $\qquad U_+ \cap U_- = \mathbb{R}^3 \setminus \underbrace{\{(0,y) \mid y \in \mathbb{R}\}}_{\text{Dirac string}}$

$A_+$ on $U_+$ and $A_-$ on $U_-$

such that $B = dA_+$ on $U_+$ and $B = dA_-$ on $U_-$

also want $A_+ = A_- + i g^{-1} dg$ $\quad g : U_+ \cap U_- \to U(1)$

$$A_+ = \alpha\, dx + \beta\, dy + \gamma\, dz$$

$$B = \frac{q}{4\pi \|\vec{x}\|^3} \left( x\, dy \wedge dz + y\, dz \wedge dx + z\, dx \wedge dy \right)$$

$$dA_+ = \partial_y \alpha\, dy \wedge dx + \partial_z \alpha\, dz \wedge dx + \partial_x \beta\, dx \wedge dy + \partial_z \beta\, dz \wedge dy + \partial_x \gamma\, dx \wedge dz + \partial_y \gamma\, dy \wedge dz$$

$$\frac{q}{4\pi \|x\|^3} x = \partial_y \gamma - \partial_z \beta$$

$$\frac{q}{4\pi \|x\|^3} y = \partial_z \alpha - \partial_x \gamma$$

$$\frac{q}{4\pi \|x\|^3} z = \partial_x \beta - \partial_y \alpha$$